MOVIE STREAMING PLATFORM

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PROBLEM STATEMENT

In movie streaming platforms, content is divided into many genres and subgenres.

PROBLEMS:

- Difficult to manage large number of genres manually
- No clear hierarchical structure
- Hard to search or update categories

SOLUTION:

Design a system that:

- Organizes data hierarchically
- Supports dynamic operations (CRUD)
- Provides easy navigation

REAL WORLD APPLICATION:

This system is used in:

- Streaming platforms (genre classification)
- Music apps (genres & playlists)
- Content management systems

Benefits:

- Easy navigation
- Better organization
- Faster searching

DATA STRUCTERS USED

WE USED A TREE DATA STRUCTURE (CHILD-SIBLING REPRESENTATION)

Each node contains:

- Name of genre/subgenre
- Pointer to child
- Pointer to sibling

Why Tree?

- Represents hierarchy naturally
- Supports multiple levels
- Easy to manage relationships

SYSTEM DESIGN

System includes:

- Menu-driven program
- Tree structure with nodes
- Each node contains:
- Name
- Child pointer
- Sibling pointer

Operations:

- Add Genre
- Add Subgenre
- Update
- Delete
- Search
- Display

ALGORITHM

1. Start program
2. Initialize root as NULL
3. Display menu
4. Take user choice
5. Perform operation:
   - Insert (add genre/subgenre)
   - Update node
   - Delete node
   - Search node
   - Display tree
6. Repeat until user exits

IMPLEMENTATION

language used:C programming

Main functions:

- addGenre()
- addSubgenre()
- updateNode()
- deleteNode()
- search()
- displayTree()

Special feature:

- Tree shown in structured format (round-style nodes)

DEMO OUTPUT

ADDING GENRE:

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 1
Enter Genre name: Comedy
```

DEMO OUTPUT

ADDING SUBGENRES:

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 2
Enter parent name: Comedy
Enter subgenre name: Rom-Com
```

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 2
Enter parent name: Comedy
Enter subgenre name: Dark
```

DEMO OUTPUT

ADDING SUBGENRES:

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 2
Enter parent name: Comedy
Enter subgenre name: Horror
```

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 2
Enter parent name: Comedy
Enter subgenre name: Action
```

DEMO OUTPUT

UPDATE:

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 3
Enter old name: Action
Enter new name: StandUp
```

DELETE:

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 4
Enter parent name: Comedy
Enter node to delete: Horror
Deleted successfully
```

DEMO OUTPUT

SEARCH:

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 5
Enter name to search: Rom-Com
Found!
```

DISPLAY:

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 6

--- Genre Tree ---

( Comedy )
   |
   v
    ( Rom-Com )
    ( Dark )
    ( StandUp )
```

DEMO OUTPUT

RESET:

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 7
Tree reset done
```

EXIT:

```
===== Movie Streaming Platform =====
1. Add Genre
2. Add Subgenre
3. Update Node
4. Delete Node
5. Search
6. Display
7. Reset Tree
8. Exit
Enter your choice: 8
Exited!
```

CONCLUSION

- Successfully implemented tree-based system
- CRUD operations working correctly
- Clear hierarchical representation
- Easy to understand and use

FUTURE IMPROVEMENTS:

- Add GUI (Graphical Interface)
- Store data in file/database
- Add movie details inside genres
- Improve display with graphics

Thank You So Much!